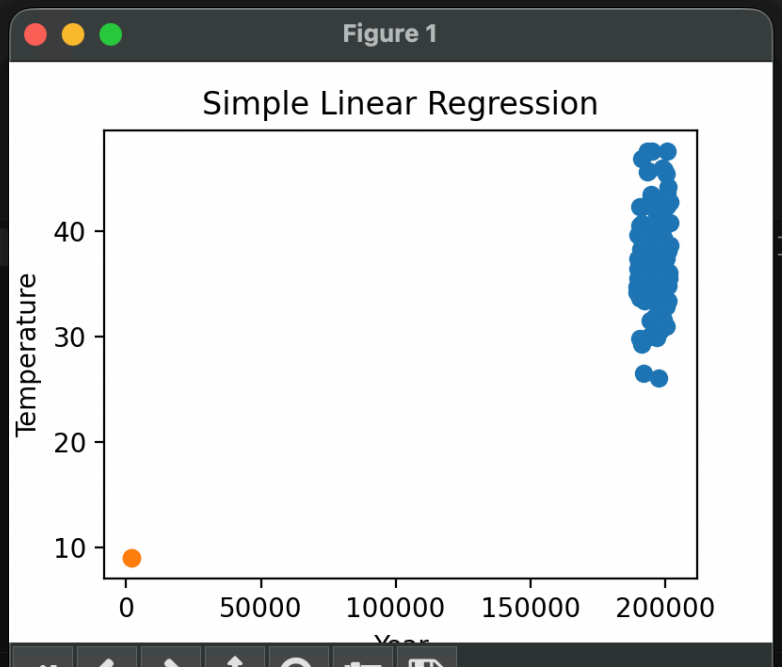
exercise1.py .../WilliamsBenjamin_HW4/... U | avgHigh_jan_1895-2018.csv .../Exercise1

HW > HW4 > WilliamsBenjamin_HW4 > Exercise1 > exercise1.py > ...

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

df = pd.read_csv("avgHigh_jan_1895-2018.csv")
X = df.iloc[:, 0].values.reshape(-1, 1)
y = df.iloc[:, 1].values

model = LinearRegression()
model.fit(X, y)

years_pred = np.array([2019, 2023, 2024]).reshape(-1, 1)
preds = model.predict(years_pred)

print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)

for yr, pr in zip(years_pred.flatten(), preds):
    print("Predicted temperature for Jan", yr, ":", pr)

plt.scatter(X, y)
plt.plot(X, model.predict(X))
plt.scatter(years_pred, preds)
plt.xlabel("Year")
plt.ylabel("Temperature")
plt.title("Simple Linear Regression")
plt.show()
```



CHAT TERMINAL

```
benwilliams@Bens-MacBook-Air-2 MachineLearning % /Users/benwilliams/D
ocuments/De
velopment/MachineLearning/.venv/bin/pyth
on /Users/benwilliams/Documents/Developm
ent/MachineLearning/HW/HW4/WilliamsBenja
min_HW4/Exercise1/exercise1.py
Traceback (most recent call last):
  File "/Users/benwilliams/Documents/Development/MachineLearning/HW/H
W4/WilliamsBenjamin_HW4/Exercise1/exercise1.py", line 6, in <module>
    df = pd.read_csv("avgHigh_jan_1895-2018.csv")
  File "/Users/benwilliams/Documents/Development/MachineLearning/.ven
v/lib/python3.13/site-packages/pandas/io/parsers/readers.py", line 87
3, in read_csv
    return _read(filepath_or_buffer, kwds)
  File "/Users/benwilliams/Documents/Development/MachineLearning/.ven
v/lib/python3.13/site-packages/pandas/io/parsers/readers.py", line 30
0, in _read
    parser = TextFileReader(filepath_or_buffer, **kwds)
  File "/Users/benwilliams/Documents/Development/MachineLearning/.ven
v/lib/python3.13/site-packages/pandas/io/parsers/readers.py", line 16
45, in __init__
    self._engine = self._make_engine(f, self.engine)
                   ~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^
  File "/Users/benwilliams/Documents/Development/MachineLearning/.ven
v/lib/python3.13/site-packages/pandas/io/parsers/readers.py", line 19
04, in _make_engine
    self.handles = get_handle(
                   ~~~~~~~~~~^
        f,
        ^^
    ...<6 lines>...
        storage_options=self.options.get("storage_options", None),
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
  File "/Users/benwilliams/Documents/Development/MachineLearning/.ven
v/lib/python3.13/site-packages/pandas/io/common.py", line 926, in get
_handle
    handle = open(
        handle,
    ...<3 lines>...
        newline="",
    )
FileNotFoundError: [Errno 2] No such file or directory: 'avgHigh_jan_
1895-2018.csv'
benwilliams@Bens-MacBook-Air-2 MachineLearning % cd HW
benwilliams@Bens-MacBook-Air-2 HW % cd H
W4
benwilliams@Bens-MacBook-Air-2 HW4 % cd
Exercise1
cd: no such file or directory: Exercise1
benwilliams@Bens-MacBook-Air-2 HW4 % ls
WilliamsBenjamin_HW4
benwilliams@Bens-MacBook-Air-2 HW4 % cd
benwilliams@Bens-MacBook-Air-2 WilliamsBenjamin_HW4 % cd Exercise1
benwilliams@Bens-MacBook-Air-2 Exercise1 % /Users/benwilliams/Documen
ts/Development/MachineLearning/.venv/bin/python /Users/benwilliams/Do
cuments/Development/MachineLearning/HW/HW4/WilliamsBenjamin_HW4/Exerc
ise1/exercise1.py
Slope: 0.00014771361132966156
Intercept: 8.694845520062973
Predicted temperature for Jan 2019 : 8.99307930133756
Predicted temperature for Jan 2023 : 8.993670155782878
Predicted temperature for Jan 2024 : 8.993817869394208
```

MachineLearning

exercise2.py .../WilliamsBenjamin_HW4/... U | avgHigh_jan_1895-2018.csv .../Exercise

HW > HW4 > WilliamsBenjamin_HW4 > Exercise2 > exercise2.py > ...

```python
import numpy as np
import pandas as pd
from sklearn.linear_model import LinearRegression

df = pd.read_csv("avgHigh_jan_1895-2018.csv")
X = df.iloc[:, 0].values.reshape(-1, 1)
y = df.iloc[:, 1].values

test_size = float(input("Enter test size (e.g., 0.2): "))
split = int(len(X) * (1 - test_size))

X_train = X[:split]
y_train = y[:split]
X_test = X[split:]
y_test = y[split:]

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

for actual, pred in zip(y_test, y_pred):
    print("Actual:", actual, "Predicted:", pred)

rmse = np.sqrt(np.mean((y_test - y_pred) ** 2))
print("RMSE:", rmse)
```

Filter | Tasks

CHAT | TERMINAL

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MachineLearning

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Exercise2
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image.png U
Exercise3
exercise3.py U
materials.csv U
Exercise4
exercise4.py U
Exercise5
exercise5.py U
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b 1
b 2
VilliamsBenjamin_Lab2
ab2.pdf
VilliamsBenjamin_Lab2.zip
b 5 / WilliamsBenjamin_Lab5
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gHigh_jan_1895-2018.csv .../Exercise2 U | exercise3.py .../WilliamsBenjamin_HW4/... U

HW > HW4 > WilliamsBenjamin_HW4 > Exercise3 > exercise3.py > ...

```python
import numpy as np
import pandas as pd

df = pd.read_csv("materials.csv")

X = df.iloc[:, :-1].values
y = df.iloc[:, -1].values
feature_names = df.columns[:-1]

for i in range(X.shape[1]):
    r = np.corrcoef(X[:, i], y)[0, 1]
    print("Correlation between", feature_names[i], "and Strength:", r)

X_design = np.c_[np.ones(len(X)), X]
beta = np.linalg.inv(X_design.T @ X_design) @ X_design.T @ y

points = np.array([
    [32.1, 37.5, 128.95],
    [36.9, 35.37, 130.03]
])

for p in points:
    x_vec = np.insert(p, 0, 1)
    pred = 0
    for i in range(len(beta)):
        pred += beta[i] * x_vec[i]
    print("Predicted Strength:", pred)
```

Filter | Tasks

CHAT | TERMINAL

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```

MachineLearning

materials.csv .../Exercise3 U | exercise4.py U | materials.csv .../Exercise4 U

HW > HW4 > WilliamsBenjamin_HW4 > Exercise4 > exercise4.py > ...

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from mpl_toolkits.mplot3d import Axes3D

df = pd.read_csv("materials.csv")

X = df.iloc[:, :-1]
y = df.iloc[:, -1]

corrs = [abs(np.corrcoef(X.iloc[:, i], y)[0, 1]) for i in range(X.shape[1])]
top2_idx = np.argsort(corrs)[-2:]

X2 = X.iloc[:, top2_idx].values

model = LinearRegression()
model.fit(X2, y)

x1 = X2[:, 0]
x2 = X2[:, 1]

x1_grid, x2_grid = np.meshgrid(
    np.linspace(x1.min(), x1.max(), 30),
    np.linspace(x2.min(), x2.max(), 30)
)

grid = np.c_[x1_grid.ravel(), x2_grid.ravel()]
y_grid = model.predict(grid).reshape(x1_grid.shape)

fig = plt.figure()
ax = fig.add_subplot(111, projection="3d")

ax.plot_surface(x1_grid, x2_grid, y_grid, alpha=0.5)
ax.scatter(x1, x2, y)
```

Filter | Tasks

CHAT | TERMINAL

```
_handle
    handle = open(
        handle,
```

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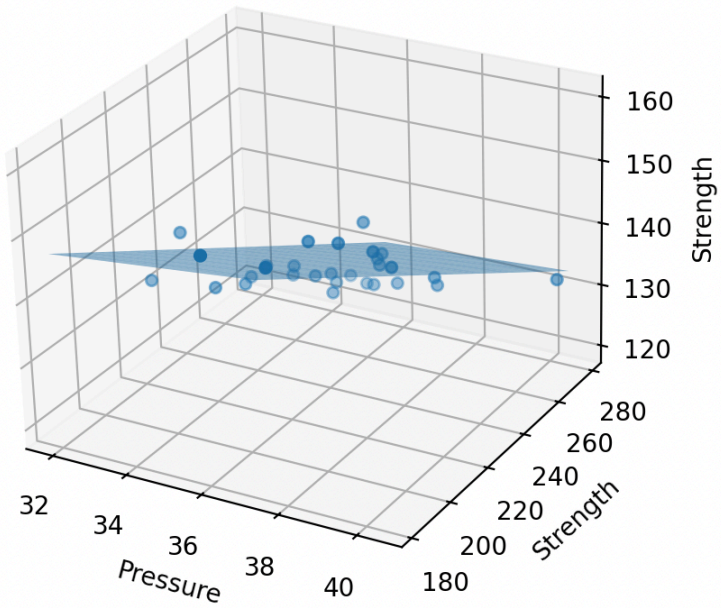
Figure 1




```
    )
FileNotFoundError: [Errno 2] No such file or directory: 'materials.cs
v'
benwilliams@Bens-MacBook-Air-2 Exercise4 % /Users/benwilliams/Documen
ts/Development/MachineLearning/.venv/bin/python /Users/benwilliams/Do
cuments/Development/MachineLearning/HW/HW4/WilliamsBenjamin_HW4/Exerc
ise4/exercise4.py
benwilliams@Bens-MacBook-Air-2 Exercise4 % /Users/benwilliams/Documen
ts/Development/MachineLearning/.venv/bin/python /Users/benwilliams/Do
cuments/Development/MachineLearning/HW/HW4/WilliamsBenjamin_HW4/Exerc
ise4/exercise4.py
benwilliams@Bens-MacBook-Air-2 Exercise4 % /Users/benwilliams/Documen
ts/Development/MachineLearning/.venv/bin/python /Users/benwilliams/Do
cuments/Development/MachineLearning/HW/HW4/WilliamsBenjamin_HW4/Exerc
ise4/exercise4.py
```

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MachineLearning

materials.csv .../Exercise4 U | exercise5.py U | materialsOutliers.csv U

HW > HW4 > WilliamsBenjamin_HW4 > Exercise5 > exercise5.py > ...

```python
import numpy as np
import pandas as pd
from sklearn.linear_model import RANSACRegressor, LinearRegression

df = pd.read_csv("materialsOutliers.csv")

X = df.iloc[:, :-1].values
y = df.iloc[:, -1].values

mask = np.ones(len(df), dtype=bool)

for i in range(X.shape[1]):
    ransac = RANSACRegressor(
        estimator=LinearRegression(),
        residual_threshold=15,
        stop_probability=1.0
    )
    ransac.fit(X[:, i].reshape(-1, 1), y)
    mask = mask & ransac.inlier_mask_

X_clean = X[mask]
y_clean = y[mask]

model = LinearRegression()
model.fit(X_clean, y_clean)

print("Final Coefficients:", model.coef_)
print("Intercept:", model.intercept_)
```

Filter | Tasks

CHAT TERMINAL

```
v'
benwilliams@Bens-MacBook-Air-2 Exercise4 % /Users/benwilliams/Documen
ts/Development/MachineLearning/.venv/bin/python /Users/benwilliams/Do
cuments/Development/MachineLearning/HW/HW4/WilliamsBenjamin_HW4/Exerc
ise4/exercise4.py
benwilliams@Bens-MacBook-Air-2 Exercise4 % /Users/benwilliams/Documen
ts/Development/MachineLearning/.venv/bin/python /Users/benwilliams/Do
cuments/Development/MachineLearning/HW/HW4/WilliamsBenjamin_HW4/Exerc
ise4/exercise4.py
benwilliams@Bens-MacBook-Air-2 Exercise4 % /Users/benwilliams/Documen
ts/Development/MachineLearning/.venv/bin/python /Users/benwilliams/Do
cuments/Development/MachineLearning/HW/HW4/WilliamsBenjamin_HW4/Exerc
ise4/exercise4.py
cd ../E
benwilliams@Bens-MacBook-Air-2 Exercise4 % cd ../Exercise5
benwilliams@Bens-MacBook-Air-2 Exercise5 % /Users/benwilliams/Documen
ts/Development/MachineLearning/.venv/bin/python /Users/benwilliams/Do
cuments/Development/MachineLearning/HW/HW4/WilliamsBenjamin_HW4/Exerc
ise5/exercise5.py
Traceback (most recent call last):
  File "/Users/benwilliams/Documents/Development/MachineLearning/HW/H
W4/WilliamsBenjamin_HW4/Exercise5/exercise5.py", line 5, in <module>
    df = pd.read_csv("materialsOutliers.csv")
  File "/Users/benwilliams/Documents/Development/MachineLearning/.ven
v/lib/python3.13/site-packages/pandas/io/parsers/readers.py", line 87
3, in read_csv
    return _read(filepath_or_buffer, kwds)
  File "/Users/benwilliams/Documents/Development/MachineLearning/.ven
v/lib/python3.13/site-packages/pandas/io/parsers/readers.py", line 30
0, in _read
    parser = TextFileReader(filepath_or_buffer, **kwds)
  File "/Users/benwilliams/Documents/Development/MachineLearning/.ven
v/lib/python3.13/site-packages/pandas/io/parsers/readers.py", line 16
45, in __init__
    self._engine = self._make_engine(f, self.engine)
                   ~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^
  File "/Users/benwilliams/Documents/Development/MachineLearning/.ven
v/lib/python3.13/site-packages/pandas/io/parsers/readers.py", line 19
04, in _make_engine
    self.handles = get_handle(
                   ~~~~~~~~~~^
        f,
        ^^
    ...<6 lines>...
        storage_options=self.options.get("storage_options", None),
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
  File "/Users/benwilliams/Documents/Development/MachineLearning/.ven
v/lib/python3.13/site-packages/pandas/io/common.py", line 926, in get
_handle
    handle = open(
        handle,
    ...<3 lines>...
        newline="",
    )
FileNotFoundError: [Errno 2] No such file or directory: 'materialsOut
liers.csv'
benwilliams@Bens-MacBook-Air-2 Exercise5 % /Users/benwilliams/Documen
ts/Development/MachineLearning/.venv/bin/python /Users/benwilliams/Do
cuments/Development/MachineLearning/HW/HW4/WilliamsBenjamin_HW4/Exerc
ise5/exercise5.py
Final Coefficients: [-0.26023958  0.05335903  0.18666974]
Intercept: 189.89442410629354
benwilliams@Bens-MacBook-Air-2 Exercise5 %
```